

Girijananda Chowdhury University
Department of Mechanical Engineering
Solid Mechanics (Civil Engineering)
B.Tech 3rd Semester (Class Test)

Time: 1 Hour

Total Marks: 30

1. Derive the expression $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$. 10
2. State the assumptions made in the theory of torsion 5
3. An electric rotates at 200rpm and receives 300KW from the driving engine. The armature of the generator is 60cm long and located between bearings 120cm center to center. Owing to the combine weight of armature and the magnetic pull, the shaft is subjected to 9000kg acting at right angles to the shaft. The allowable tensile stress and shear stress of the shaft is 74.66N/mm² and shear stress is 65.33N/mm² respectively. Find the diameter of the shaft. 10
4. A cylindrical pipe of diameter 1.5m and the thickness 1.5 cm is subjected to an internal fluid pressure of 1.2N/mm². Determine (i) Longitudinal stress developed in the pipe, and (ii) circumferential stress developed in the pipe. 5